Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

INSTALLER GUIDE · BATTERY COMMUNICATION

Pylontech: parallel battery communication

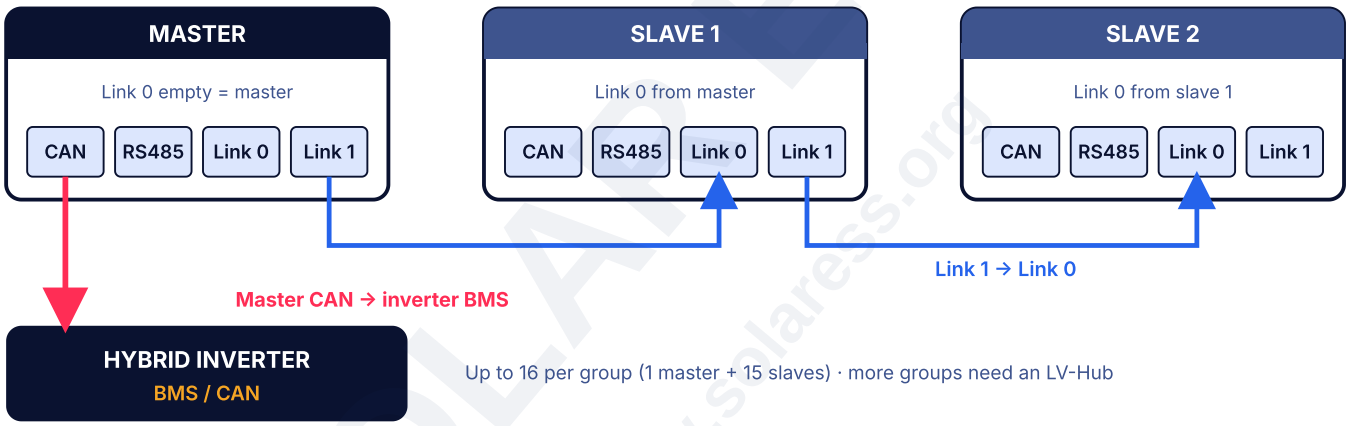
US5000 · US3000C · US2000C low-voltage 48 V modules (one group, up to 16 batteries) · note for Force-L2

One rule for every brand: only the **master** battery talks to the inverter. All other batteries are chained to it.

1 · Set the master / address

The master is set by the cable, not a switch. The master is the battery whose **Link Port 0 is empty**. Leave the ADD switch at factory default for a single group: Dip2 keeps the CAN terminal resistor on the master. Dip1 sets RS485 speed (only change it if the inverter needs 9600 bps). Restart the batteries after changing any DIP. In a mixed set, use the newest model (US5000) as master.

2 · Connect the communication cables



1. Inverter **BMS** port → **Master CAN** (or **RS485** if the inverter uses RS485).
2. **Master Link Port 1** → **Slave 1 Link Port 0**, then Slave 1 Link Port 1 → Slave 2 Link Port 0, and so on.
3. CAN/RS485 port pinout: CAN-H pin 4, CAN-L pin 5, CAN-GND pin 6, RS485 A/B pins 7/8. Pin 1 must stay unconnected. Use the Pylontech inverter cable where supplied.
4. Power: each battery has two pairs of terminals rated 100 A each. Use enough pairs for the system current and fit a breaker to the inverter.

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3 · Inverter settings

- **Solis:** Battery Select → **Pylon (US5000)**, CAN.
- **Deye:** Battery Mode = Lithium, lithium protocol **00** (Pylon CAN). Tick Activate and BMS_Err_Stop.
- **GoodWe:** select Pylontech in SolarGo, CAN (inverter firmware V14 or later).
- **Growatt SPF:** RS485 at 9600 bps, so set battery Dip1 = 1 and restart. Growatt SPH: CAN.
- For more than one group, set the masters as the LV-Hub manual describes.

4 · Power-on sequence

1. Check all power and communication cables.
2. Switch on the external DC breaker (US5000-B: each module's breaker too).
3. Switch on the power switch of every module.
4. Press the red **SW / START** button on the **master** for more than 0.5 s. LEDs light up battery by battery from the master (about 3 s).
5. Start the inverter. Switch off: remove the charge source, press SW on the master, then turn off the module switches.

If modules are at different charge levels, leave the system idle for at least 15 minutes, or until the SOC LEDs differ by no more than one dot, before loading it.

5 · Cable & good practice

Custom cable	Wire only pins 4/5/6 (CAN) or 7/8 (RS485); leave all other pins open.
Force-L2	No DIP: stacks link Link Port 1 → Link Port 0 (≤ 2 m), stack 1 is the master and goes to the inverter.
Cable	Standard straight LAN cable (CAT5e/CAT6, T568B both ends). No crossover.
Routing	Keep communication cables away from DC power cables. Plugs must click in fully.
Mixing	Never mix brands, models or capacities in one battery bank or one communication chain.
Voltage	Before paralleling, charge or rest all packs so their voltages are close.

6 · Troubleshooting: "no SOC" or "BMS communication fail"

Inverter "Batt_Comm_FAIL"	Usually the wrong cable: use the Pylontech CAN cable or re-pin to 4/5.
Inverter sees only one battery	A Link cable is missing, or a cable is in Link Port 0 of the master.
ALM flashing / ALM solid	Alarm / protection active: check voltage, current and temperature.
Buzzer with solid ALM	Reversed cable or MOSFET fault: switch everything off and check the wiring.
Changed DIP has no effect	Restart all batteries after any DIP change.

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